



CDISC DATA EXCHANGE STANDARDS

DDS Profiles

One model. Many use cases. Machine-readable contracts for each.

Introducing the DDS Project Charter and DDS Profiles (DDS-P)

DDS Project | August 2026

What is the Data Definition Specification?

A machine-readable data contract: a structured, version-controlled specification for generating, transforming, and exchanging CDISC-compliant clinical research data. Expressed as a LinkML model with JSON serialization.



A data contract, not the data

Defines schema, semantics, constraints, and declarative transformations. References datasets and systems rather than containing data — enabling data virtualization and dynamic views.



Source, target, and the logic between

DDS explicitly describes source datasets, target datasets, and the transformation logic required to produce one from the other — the metadata that drives automation.



Versioned and machine-first

A LinkML model in an open GitHub repository, released incrementally as a published CDISC data exchange standard, with developers invited to apply, test, and improve it.

Positioning: DDS is planned to represent a new Define-JSON — a future submission standard to replace Define-XML v2.x.

The charter at a glance



Incremental, developer-tested releases

Develop and publish DDS incrementally as a machine-readable CDISC standard, inviting developers to apply the model to test and improve it.



Profiles guide the development

The development process is guided by DDS Profiles targeting a prioritized set of use cases — profiles are how the standard meets reality.



360i adoption from day one

The 360i project will begin using DDS to represent the metadata needed to drive automation; Define-XML and CRF/aCRF generation are already covered there.



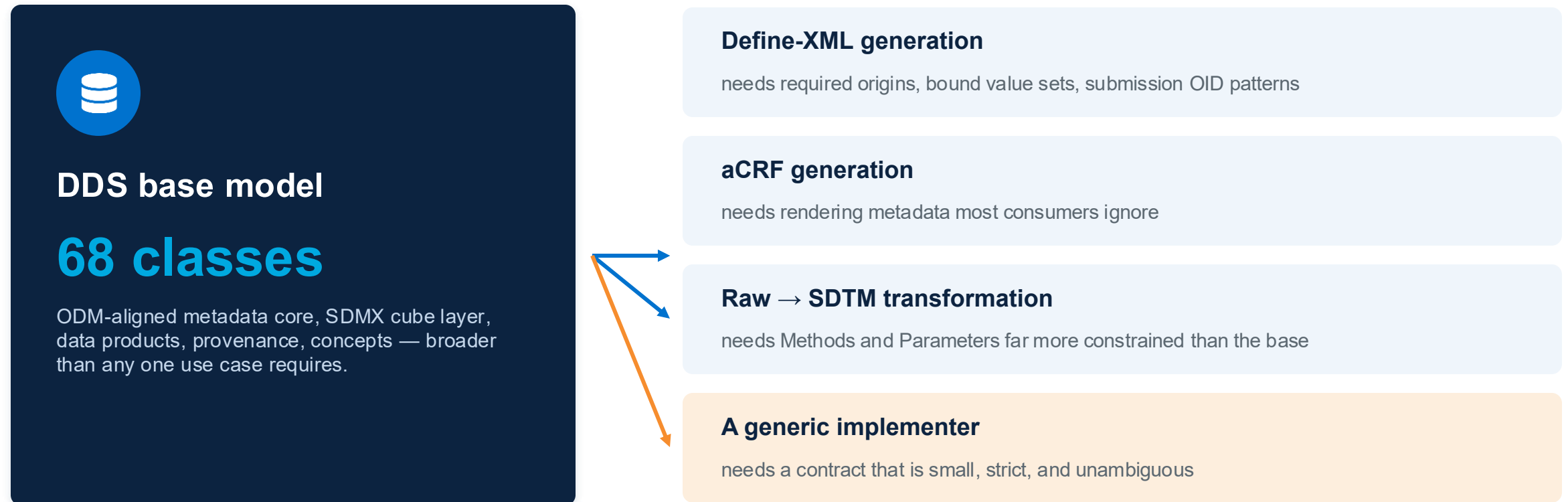
Built for generative AI

DDS-P Skills (skill.md files) will guide AI agents implementing DDS-based solutions — DDS improves the quality of AI use with CDISC standards.

Deliverables include the normative specification, LinkML model + JSON Schema, a living User Guide, four initial profiles, examples, and AI Skills.

One broad model. Very specific jobs.

DDS is deliberately broad — a Rosetta Stone across ODM, USDM, Dataset-JSON, SDMX, FHIR, and OMOP. No single implementation needs all of it.



The gap: implementers should not have to interpret the full model to serve one use case.

DDS Profiles: use-case slices of the model

“A DDS Profile carves a use-case-specific, machine-readable slice out of the base model.”

Tightens cardinality • **Narrows** value sets • **Fixes** values • **Prunes** irrelevant classes • adds **declared extensions**

A PROVEN PATTERN — TWICE OVER



HL7 FHIR profiles

FHIR constrains its resources with StructureDefinition profiles — differential + snapshot, constraint + extension. DDS applies the same concept to one broad model.



Define-XML v2.1 itself

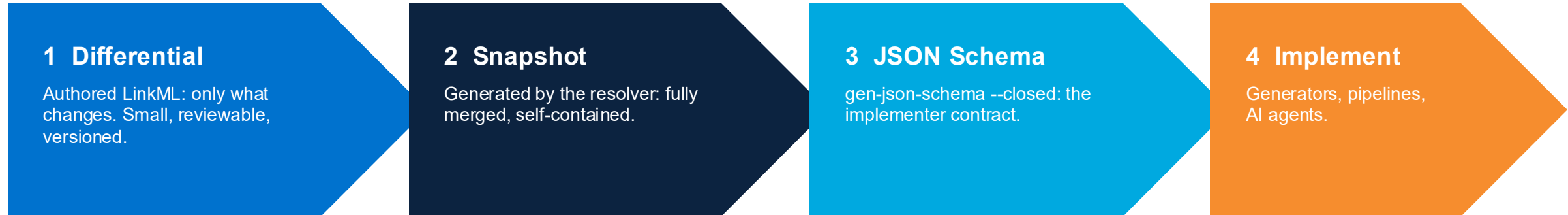
Define-XML is effectively a profile of ODM v1.3.2: it constrains and extends the ODM model to serve the submission use case, and Define-XML files must conform to its specification — exactly the relationship DDS-P instances have to their profile.



Implementers build against the profile, not the full model.

Smaller, stricter, unambiguous — DDS-Ps are normative, published as required components of DDS Implementation Guides, and can themselves be extended and constrained to meet research needs.

How a profile works: differential → snapshot → schema



Only tightening is allowed

The resolver enforces it: any loosening of the base model is a build error, not a warning.



Extensions are declared & namespaced

Strip a profile's extensions and the instance still validates against base DDS — every profiled instance stays interoperable.



Deterministic and CI-enforced

Same inputs → byte-identical snapshot. Snapshots are generated artifacts; hand edits are never permitted.

Every LinkML generator works on a snapshot: JSON Schema, Pydantic, TypeScript, OWL, docs.

Eight use cases. Four initial profiles.

1	Define-XML	INITIAL PROFILE	2	CRF / aCRF	INITIAL PROFILE
3	Data Transfer Agreement	INITIAL PROFILE	4	Raw → SDTM conversions	INITIAL PROFILE
5	Pipeline automation		6	ODM Metadata	
7	End-to-end traceability		8	RWD interoperability (FHIR, OMOP)	

Mapping is deliberate, not mechanical: some use cases map 1:1 to a profile; others — like traceability — are features woven through profiles; related use cases may share one profile.

Secondary (out of initial scope): BC conversion, BC registry/MDR, SDTM→ADaM and ADaM→ARD metaprogramming, ARD analysis dimensions, impact-analysis visualization.

The Define-XML profile runs today

Prototype profile + snapshot resolver, exercised against the LZZT reference study (SDTMIG 3.4, CDISC/NCI CT 2024-09-27).

43 / 68

classes remain after profiling — 25 pruned automatically by reachability

311

items across 21 datasets validated through the generated closed JSON Schema

4

classes of real producer defects caught in the reference instance

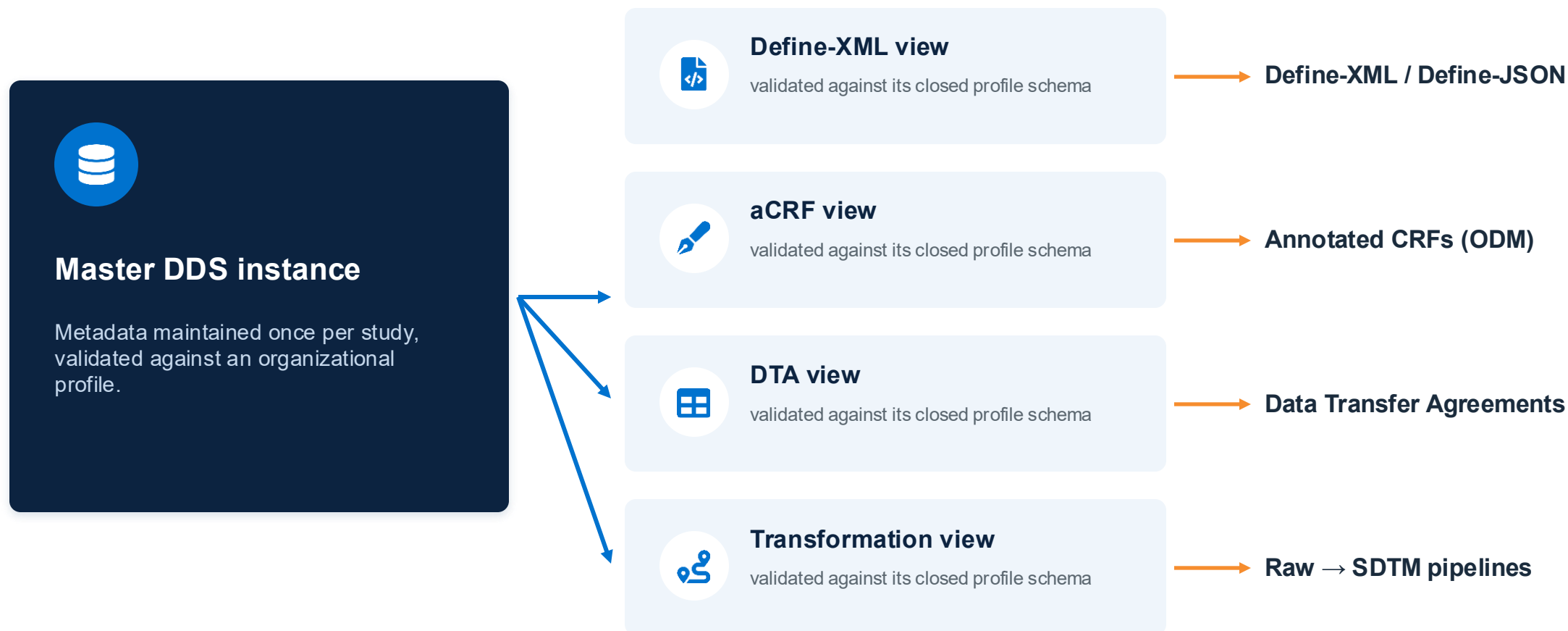
What the profile caught:

- Serialization-shape bug: origins emitted as objects where the model requires lists
- Unresolved template markers flagged against bound value sets
- null used where keys should be omitted (length)
- Unknown document keys rejected by the closed schema



Validation as a quality gate for generator output — exactly the automation the charter targets.

One study instance. Many profile views.



Profiles only tighten — so projecting a view is purely mechanical, conformance is claimed in the instance, and every pipeline stage consumes a validated contract while the organization maintains truth exactly once.

What the project will publish



1 · Specification

The normative DDS standard specification, authored in Markdown.



2 · Model & Schema

The DDS model in LinkML plus the generated DDS JSON Schema.



3 · User Guide

How to implement DDS and Profiles — a living document that grows post-publication.



4 · Four Profiles

Define-XML, CRF/aCRF, DTA, and Raw-to-SDTM — published as required components of DDS Implementation Guides.



5 · Examples

Profile example files maintained in the project GitHub repository.



6 · AI Skills

skill.md files guiding generative-AI agents through DDS-P implementations.

Profiles appear twice: as deliverable #4 and as the development method itself.

DDS defines the contract. Profiles make it implementable.

Next steps

- Review the charter draft and bring feedback to the DES team
- Join the profile working sessions — Define-XML first, CRF/aCRF next
- Developers: clone the repo and try the model against your use case



[github.com/cdisc-org/
DataExchange-DDS](https://github.com/cdisc-org/DataExchange-DDS)

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